

of the test places no constraints on doxastic states. A test resulting in a determination of $\Diamond H_0 \wedge \Diamond H_1$ is, in this sense, merely *indecisive*. Only when $\Vdash \Box H_i$ would any constraints on doxis be imposed. Naturally, in the design of the experiment we would ideally like to assess the probability that the experiment will be indecisive, that is that $\Diamond H_0 \wedge \Diamond H_1$. This is closely related to the decisive power function.

Definition 4.6. The indecisiveness function $i_\Diamond(\theta, n)$ is the probability of yielding $\Diamond H_0 \wedge \Diamond H_1$ given θ and sample size n :

$$\begin{aligned} i_\Diamond(\theta, n) &= \mathbb{P}_\theta(\Diamond H_0 \wedge \Diamond H_1) \\ (21) \quad &= 1 - \mathbb{P}_\theta(\Box H_0 \vee \Box H_1) \\ &= 1 - \delta(\theta, n). \end{aligned}$$

In the two-sided case, the indecisiveness of the test is precisely the rate of Type II error since the power function and the decisive power function coincide.

A more theoretical example illustrates how the topological properties of H_0 relate to indecisiveness.

Proposition 4.7. Let $H_0 \subset \Theta$ be dense-codense. Then $i_\Diamond(\theta, n) = 1$.

Proof. Since H_0 is dense/codense, if $U \subset \Theta$ is open nonempty then $H_0 \cap U$ and $H_1 \cap U$ are nonempty. Since confidence intervals are open nonempty, $\Vdash \Diamond H_0 \wedge \Diamond H_1$, so $i_\Diamond(\theta, n) = 1$. $\square$

5. EQUIVALENCE TESTING IS A MODEL OF MAYO'S SEVERE TESTING

5.1. Two One-Sided Test on Loss Functions are a Confirmation Logic of Empirical Adequacy. By a predictive theory τ , I mean a theory which makes predictions—either deterministic or probabilistic—of the outcomes of events under a certain experimental context. That is, given some experimental setup X , there is a function f_τ which generates predictions $f_\tau(X)$. Let $A(X)$ be the actual outcome of the experimental setup. We assume that the empirical adequacy of the theory τ is assessed according to a loss function L measuring the discrepancy between the predictions of τ and the actual outcomes: $L(f_\tau(X), A(X)) \in \mathbb{R}^{\geq 0}$. Our strategy for confirming τ relative to the experiment X is to estimate and bound a statistic $\theta = \theta(L, \tau)$ of the loss function.

Given a tolerance for error—an *adequacy margin*— $M > 0$, we can perform the following test on θ at level α :

$$(22) \quad H_0 : |\theta| \leq M \text{ vs. } H_1 : |\theta| > M.$$

Formally, this test is the test of TOST with the null and alternative hypotheses inverted. We interpret the null H_0 as expressing the empirical adequacy of the theory τ up to level α . As discussed in 3.5, a hypothesis H_0 with nonempty interior admits the possibility of confirmation up to level α . Thus, relative to an experimental procedure X , loss function L , and adequacy margin M there is a well-defined sense in which we can say that the evidence E *statistically confirms* the empirical adequacy of the theory τ . I want to emphasize, as Mayo does, that this is relative to the parameters X , L , and M : I am not claiming that $\Box H_0$ confirms τ ; only the empirical adequacy of τ with respect to the experiment X is being confirmed.

5.2. Two One-Sided Testing Satisfies the Desiderata for Mayo's Severe Testing. Mayo and other error statisticians instead advocate for a modern recasting of NHST—severe testing—as the appropriate framework guiding the use of statistical methods. Central to the error statistician is the question:

When do data E provide good evidence for/a good test of hypothesis τ ?

The error statistician will invoke some form of a **Severity Principle** to answer this question:

(Weak Severity Principle) Data E **does not** provide good evidence for τ if E is the result of a **test procedure** T with very low probability of uncovering the falsity of τ [9] (Page 21).

A converse is given by:

(Full Severity Principle) Data E provides good evidence for τ to the extent that test T has been **severely passed** by τ [9] (Page 21).

The error statistician naturally asks *which* hypotheses are amenable to error-theoretic analysis. This question is of utmost importance as the Full Severity Principle suggests the following account of scientific content: the hypotheses τ that have scientific content are precisely those which are severely testable.

The notion of severe testing as described by Mayo is defined as follows:

Definition 5.1. A hypothesis τ passes a severe test relative to experiment X with data E if (and only if):

- (1) E agrees with or “fits” τ (for a suitable notion of fit), and
- (2) experiment X would (with very high probability) have produced a result that fits τ less well than E does, if τ were false or incorrect. [7] (Page 99)

I propose the following model of severe testing. Let τ be a predictive theory in the sense described in the prior section. The predictions of τ can be assessed by a loss function $L(\cdot, \cdot)$, and margin of adequacy M . The *empirical adequacy hypothesis* of τ relative to L and M is: $H_\tau : |\theta_L| \leq M$. I claim that if the experiment yields $\models \Box H_\tau$, we may interpret that as saying that τ has passed a severe test (to the extent that the loss function L and adequacy margin M are acceptably stringent).

First, we argue that $\Box H_\tau$ entails that E fits with τ . This is immediate: a truth condition for $\models H_0$ is that the confidence interval loss statistic θ_L is wholly contained within the margin of adequacy interval $[-M, M]$.

Second, we argue that this procedure validates the second clause. Reordering the clause, it reads:

$$\begin{array}{l}
 (23) \quad \underbrace{\tau \text{ false or incorrect}}_{(a)} \Rightarrow \\
 \underbrace{\text{The probability that the experiment would have produced a result more discrepant with } \tau \text{ is very high.}}_{(b)}
 \end{array}$$

In this context, I interpret (a) as meaning that the loss statistic θ is not in H_0 , i.e. that $\theta \in H_1$. At first blush this may seem to be a misrendering of the intention of the clause (a). Even if so, it seems to be a necessary amendment: for any theory τ , we can construct an alternate theory τ' such that τ and τ' agree about the results of a specific experiment but disagree regarding other outcomes. Unless the experiment E were exhaustive of all of τ 's predictions, such τ' exist. Therefore, there exist nearby theories for which the probability expressed in (b) will be low—or even zero. A strict interpretation of (a) renders severe testability impossible.

Regarding clause (b), and assuming that by (a) we mean $\theta_L \in H_1$, we interpret a result more discrepant from τ than $\Box H_0$ as meaning the experiment returned $\neg \Box H_0$, that is, either $\Box H_1$ or $\Diamond H_0 \wedge \Diamond H_1$. This is a plausible interpretation of (b), as in both $\Box H_1$ or $\Diamond H_0 \wedge \Diamond H_1$ the confidence interval for θ_L intersects H_1 nontrivially and is therefore more discrepant with τ than $\Box H_1$. Moreover, since for $\theta \in H_1$ we have $\mathbb{P}_\theta(\Box H_0) \leq \alpha$,

$$\mathbb{P}_\theta(\neg \Box H_0) = \mathbb{P}_\theta(\Box H_1 \vee (\Diamond H_0 \wedge \Diamond H_1)) \geq 1 - \alpha.$$

Thus, by specifying a sufficiently small α , the probability statement second prong in the severe testing criteria can be made arbitrarily high.

6. CONCLUSION

In this paper, a modal logic of short-run frequentist confirmation was developed by leveraging the duality between hypothesis testing and statistical estimation. This formal equivalence allows one to supplant decision rules based on the p-value achieving some threshold with set-theoretic relationships

between a suitably constructed confidence region and the hypothesis in question. As developed here, the approach to the hypothesis testing grounded in properties of confidence regions enjoys many felicitous properties that avoid epistemic problems common to multiple accounts of frequentist hypothesis testing.

First, there is symmetry between the null and alternative hypotheses. There is no need to select one of the hypotheses as doxastically default or as the subject of falsificational scrutiny. Rather, both null and alternative can be assessed simultaneously in light of the evidence. This process does not require any multiple testing adjustment, since H_1 is uniquely determined by the choice of H_0 : $H_1 = H_0^c$. This symmetrization better accommodates what one might call exploratory inquiry, wherein the investigator harbors no strong attitude toward either H_0 or H_1 .

Second, the underlying logic is trivalent, with the output of a test being characterized by which of three mutually exclusive unnested modal sentences is satisfied. The values correspond to *decisive* evidence in favor of H_0 , *decisive* evidence in favor of H_1 , or *indecision* between H_0 and H_1 . This decision procedure is not infallible, but does enjoy quantifiable rates of decisive error bounded above by the level of the confidence region. With respect to the formal semantics outlined in Appendix A, Proposition A.4 formally expresses the problem of induction for this logic.

Third, the shift to set-theoretic relationships between confidence regions and hypotheses allows us to exactly characterize the confirmable hypotheses based on their underlying topological structure. We found that standard two-sided hypotheses are unconfirmable but that equivalence and noninferiority tests are confirmable. This provides formal justification for Lakens' claim that two-sided hypotheses are not confirmable.

This has broad implications for scientific and statistical practice. It suggests that two-sided tests can be useful for questions of refutation or falsification, but *not* for confirmation. Therefore, strategies for the statistical confirmation of theories must be developed. In the final section of this paper, one such strategy was developed and was argued to be a model of Mayo's desiderata for severe testing. To circumvent the unconfirmability of two-sided hypotheses, the empirical adequacy of a theory τ with respect to a loss function L and adequacy margin M is subjected to a Two One-Sided Test of the hypothesis $H_\tau : |\theta_L| \leq M$. If H_τ is confirmed, then τ has passed a severe test. Crucially, this test requires a specification of both the loss function L and adequacy margin M . The suitability of the choice of these parameters is ultimately normative, and these choices should be justified by the investigator.